



The ginseng's fireiness is associated with the lowering activity of liver $\text{Na}^+ - \text{K}^+ - \text{ATPase}$



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ABSTRACT

Ethnopharmacological relevance: Ginseng is an herbal medicine used worldwide that possesses a wide range of pharmacological activities. However, its side effects are rarely discussed. The experience of Chinese medicine has revealed that taking ginseng at a high dose chronically can cause *fireiness*, i.e., the ginseng-abuse syndrome. Here, we explored the mechanism of ginseng's *fireiness* by comparing the energy metabolism of mice affected by red ginseng (RG), ginseng (GS), ginseng leaves (GL) and American ginseng (AG), which exhibit different drug properties according to the theory of TCM.

Materials and methods: KM mice were randomly divided into five groups ($n \geq 30$ per group) and administered distilled water or drugs, respectively. Mice receiving RG, GS, or GL received 4.5 g/(kg day), while the mice receiving AG received 3 g/(kg day). Control mice received distilled water. The duration of exposure for all groups was 31 days. The mice's physical characteristics, such as eye condition, rectal temperature, saliva secretion, urine, stool weight, blood coagulation time and swimming time, were measured at different times after administration. Energy metabolism indexes were measured via TSE phenoMaster/LabMaster animal monitoring system, including the mice' 24 h oxygen consumption (VO_2), carbon dioxide production (VCO_2), heat production (H) and energy expenditure (EE). Biochemical indices were measured by ultraviolet spectrophotometer and microplate reader, including pyruvic acid content in serum and succinate dehydrogenase (SDH) activity, lactate dehydrogenase (LDH) activity, the $\text{Na}^+ - \text{K}^+ - \text{ATPase}$ activity and the content of glycogen in the liver tissue.

Results: After 31 days of drug administration, mice in the RG and GS groups exhibited obviously more eye secretions, less saliva secretion and less urine. Compared with the control group, the swimming times of mice in the GS, AG and GL groups were significantly prolonged; the clotting time of mice in the GL was extended significantly; VCO_2 , H and EE of mice in the GS group were obviously increased; Pyruvate content of mice in the RG group showed an initial decrease followed by an increase; SDH activity of mice in the AG and GL groups was significantly inhibited; LDH activity of the mice showed no significant difference among different groups; $\text{Na}^+ - \text{K}^+ - \text{ATPase}$ activity of the RG and GS groups showed up-regulation initially and then down-regulation; the content of hepatic glycogen of mice in the GS and GL groups increased significantly.

Conclusion: The results demonstrated that RG and GS with their warm drug nature could enhance the body's energy metabolism to produce their *dryness* to the body. The liver $\text{Na}^+ - \text{K}^+ - \text{ATPase}$ enzyme activity may be the primary index for indicating the *fireiness* of ginseng. In addition, our results demonstrated that ginseng, especially red ginseng, is not suitable for long time application with a higher dose.

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1. Introduction

Ginseng, the dried root and rhizome of *Panax ginseng* C. A. Meyer, is a famous drug widely used for thousands of years throughout the world. The Chinese regarded it as a tonic for the restoration of strength, and so a panacea name was given to it (hence, the genus *Panax*, meaning all healing) (Siegel, 1979). Modern experiments have also indicated that ginseng possesses various pharmacological activities (Takagi et al., 1972). The long-

Abbreviations: AG, American ginseng; GS, ginseng; RG, red ginseng; GL, ginseng leaves; VO_2 , oxygen consumption; VCO_2 , carbon dioxide production; H, heat production; EE, energy expenditure; SDH, succinate dehydrogenase; LDH, lactate dehydrogenase

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term application of ginseng in Chinese medicine has found that if it is used chronically or in higher doses, it can cause *fireness* (火, in Chinese), i.e. ginseng-abused syndrome. Research has indicated that the side effects of palpitations, sleeplessness and dizziness seen with ginseng may be associated with the excitation of the central nervous system, as well as a cardiostimulant action (Wu, 1981). Chinese clinical experiments have indicated that patients appeared with symptoms of dizziness, palpitations, blurred vision, dry heat, and body burnout after receiving RG 15 g as a one-time dose (Wu, 1981). In Chinese medicine, the highest dose of ginseng that has been given is up to 50 g. The clinical observation showed that large doses of ginseng could induce insomnia, depression, and nervous disorders (Keys, 1976). To confirm the side-effects of ginseng, a rigid clinical experiment was performed that indicated that 14 of 133 long-term ginseng users (10%) exhibited side-effects. Patients who ingested an average daily dose of 3 g experienced hypertension, nervousness, sleeplessness, skin eruptions and morning diarrhea; some subjects also became euphoric and agitated. Doses of 15 g were associated with depersonalisation and confusion, while depression was reported after more than 15 g of ginseng per day (Siegel, 1979). As usual, the theory of traditional Chinese medicine (TCM) is that the ginseng *fireness*, i.e., ginseng-abuse syndrome, is caused by its *dryness* (燥性 in Chinese), which was caused by its mild property (Su and Sun, 2015). The property theory of TCM mainly includes four *Qi* (in Chinese), five flavors, and channel tropism, etc. The four *Qi* referred to are *cold*, *cool*, *mild* and *warm*. Recently, a new theory has been presented that the four *Qi* of Chinese drugs is mostly associated with energy metabolism (Huang et al., 2011). In addition, pharmacological testing has also revealed that the stimulant effects of ginseng, which provide the anti-fatigue and enhanced performance effects, may be associated with the alteration of carbohydrate metabolism and increased synthesis of glycogen and phosphorous compounds induced by ginseng (Brekham and Dardymov, 1969). These studies led us to study the *fireness* of ginseng. According to the Chinese Pharmacopoeia, red ginseng exhibits the warm property, ginseng or dried ginseng shows the mild property, ginseng leaf presents the cold property and American ginseng shows the cool property.

In addition, these four medicinal materials contain similar ginsenosides, but the proportions of main individual ginsenosides between the materials are different. Typically, ginsenosides can be classified into protopanaxadiol (PPD), protopanaxatriol (PPT) and oleanolic acid types according to their aglycone. The PPD and PPT are the predominant ginsenosides in the four drugs. Ginsenoside-Re, derived from the PPT-type, is the highest in ginseng leaves and ginsenoside-Rb₁, derived from PPD-type, is the highest in American ginseng. The total content of ginsenoside-Re and -Rg₁ derived from the PPT-type is almost equal to -Rb₁. The ginseng root contains all of these ginsenosides in higher concentrations. Other ginsenosides, such as ginsenoside-Rh₁, -Rh₂, -Rg₃, -Rg₅ and -F₄, formed either from glycosyl elimination or oxidation in the side-chain of ginsenoside-Rb₁ or -Re during the steaming process for RG by heating, are higher in the RG (Ma et al., 1996). In traditional Chinese medicine, the usage for RG and GS was different. GS was mainly used for its tonic action, such as anti-fatigue, but RG was mainly used for frail or asthenic patients. Present research indicates that RG is stronger than GS in anticancer actions (Zheng et al., 2012). Previously, the characteristic constituents of ginseng and American ginseng, as well as the activities of ginsenosides were engaged, indicating that the individual ginsenosides exhibited different activities (Dou et al., 1998; Xiang et al., 2013a, 2013b; Zhang et al., 2013). Thus, all of the ginsenosides together, in different proportions, could embody different actions. As the primary distinction between the four drugs in TCM is the difference of their properties, the comparison of their effects on energy metabolism could reflect a difference in their properties. In

general, the side effects of ginseng that occurred in some individuals included high blood pressure, insomnia, restlessness, anxiety, euphoria, diarrhea, vomiting, headache, nosebleeds, breast pain and vaginal bleeding (Jia et al., 2009).

Generally, the ancients used other drugs along with ginseng to counteract ginseng's *fireness*, called a composite prescription. However, currently the gardenized ginseng is only used for food, which has been approved by Chinese government. Thus, the research into ginseng's *fireness* is significant and imperative. However, there is currently very little research on the topic. In this paper, we took GS, RG, GL and AG as examples to study the relationship between each ginseng's *fireness* and its action on energy metabolism to reveal its mechanism of action. First, the appropriate dosages of GS, RG, GL and AG were determined by converting the doses for humans listed in the Chinese Pharmacopoeia to doses for mice based on body surface area (BSA). This paper addresses the time-dependent effects of the four drugs on energy metabolism.

2. Materials and methods

2.1. Materials and reagents

Red ginseng (Batch No. 20131009) was purchased from Dalian, China, in October 2013. Ginseng (Batch No. 20131010) was provided by Benxi Ginseng Cultivation Farm, Liaoning province of China. Ginseng leaves (Batch No. 20141012) were obtained from Kuan Dian, Liaoning province of China and American ginseng (Batch No. 20141015) was purchased from a pharmacy store in Dalian, Liaoning Province of China. All of the samples were identified by Professor Wang Bing (Liaoning University of Traditional Chinese Medicine) as the dried roots and leaves of *Panax ginseng* C. A. Meyer, as well as the dried root of *Panax quinquefolium* L, respectively. All of the voucher specimens were deposited in the herbarium and analysed to meet with the standards of China's Pharmacopoeia. Assays were performed using the protein quantitative assay kit (Batch No. 20140829, Nanjing Jian Cheng Bioengineering Institute), glycogen assay kit (Batch No. 20140901, Nanjing Jian Cheng Bioengineering Institute), SDH kit (Batch No. 20140903, Nanjing Jian Cheng Bioengineering Institute), LDH kit (Batch No. 20140829, Nanjing Jian Cheng Bioengineering Institute), pyruvate assay kit (Batch No. 20140903, Nanjing Jian Cheng Bioengineering Institute), and Na⁺-K⁺-ATPase assay kit (Batch No. 20140903, Nanjing Jian Cheng Bioengineering Institute).

2.2. Sample preparation

The mice were administered crude drug powder because there is no extraction procedure. The procedure for sample preparation was as follows: GS, GL and AG were crude drug samples, but RG is a processed form of GS. The samples of RG, GS, GL and AG were heat-dried at 50 °C, pulverized to pass through a 150-mesh sieve and then the powders were dissolved in distilled water for administration. Owing to the fact that GL is all leaf materials, it is lighter than RG, GS and AG. Additionally, triple the BSA-adjusted reference dose of GL was too thick to conduct intragastric administration. Therefore, 1.5 times the reference doses of RG, GS, GL and AG were prepared and administered twice a day. All of the drugs were given at the concentration of 0.4 ml/20 g. One and a half times the crude drug reference doses of RG, GS, and GL was 2.25 g/kg (0.045 g/20 g), while 1.5 times the crude drug reference dose of AG was 1.5 g/kg (0.03 g/20 g). Therefore, the concentration of RG, GS, and GL was 0.1125 g/ml, and the concentration of AG was 0.075 g/ml. The doses of RG, GS, GL and AG were prepared each day for administration.

2.3. Doses of four drugs

On the basis of reference, the calculation of dosages depends on the equivalent human-to-mouse dose ratio according to BSA. The calculation is $D_B \text{ (g/kg)} = D_A \times \frac{R_B}{R_A} \times \left(\frac{W_A}{W_B}\right)^{1/3}$. D_B is the dosage of mouse we want to know, D_A is the dosage for human, and W_A and W_B are the weights of people and mouse, respectively. R_A and R_B are the constant (body coefficient) of people and mouse, respectively. $D_B \text{ (g/kg)} = D_A \times \frac{0.06}{0.1} \times \left(\frac{70}{0.02}\right)^{1/3}$, $D_B \text{ (g/kg)} = D_A \times 9.1$. Therefore, the equivalent dose ratio between people and mouse is approximately 9.1, so we used 10 (Xu et al., 2002).

The doses of RG, GS and GL for humans are 9 g/d, while the daily dose of AG for humans is 6 g, according to the 2015 edition of the Chinese Pharmacopoeia. The calculated procedure as following: The total daily administration dose of RG, GS, and GL for mice was calculated as $9/60 \times 10 = 1.5 \text{ g/kg}$, while the total daily administration dose of AG for mice was calculated as $6/60 \times 10 = 1.0 \text{ g/kg}$. To choose the appropriate administration dosage, two dosages (one and three times the reference calculation) were administered in the preliminary experiment (see the Supplementary Materials). Based on the preliminary experiment, 3 times the reference dose was selected. The equivalent triple dose for mice for RG, GS and GL were calculated as 4.5 g/(kg day) while AG was 3 g/(kg day) . Because GL is leaf material, it is lighter than RG, GS and AG. Additionally, triple the reference dose of GL was too thick to conduct intragastric administration. Therefore, 1.5 times the reference doses of RG, GS, GL and AG were prepared and administered twice a day. All of the drugs were given at a concentration of 0.4 ml/20 g . One and half times the crude drug reference doses of RG, GS, GL was 2.25 g/kg , while 1.5 times the crude drug reference doses of AG was 1.5 g/kg . Therefore, the concentration of RG, GS and GL was 0.1125 g/ml , and the concentration of AG was 0.075 g/ml .

2.4. Animals and drug administration

KM male mice (18–22 g) were purchased from Liaoning Chang Sheng Biotechnology Co. LTD [License: SCXK (Liao) 2010–0001]. All mice were maintained with free access to food and water in plastic cages at $22 \pm 2^\circ\text{C}$, a relative humidity of 50–60%, and kept on a 12 h light/dark cycle. Animals were housed for one week prior to the experiments. The experimental protocols were approved by the Guide for the Care and Use of Laboratory Animals Published by the US National Institutes of Health (NIH Publication No.85–23, revised 1996). All studies were carried out in accordance with the ethics regulations of Liaoning University of TCM (131/2010).

The KM mice were randomly divided into five groups ($n=30$ per group) and administered intragastrically (*i.g.*) either distilled water or the corresponding drugs: the RG group, GS group, and GL group were administered *i.g.* the corresponding drug suspension of 2.25 g/kg , twice a day, for 31 days. The AG group was administered *i.g.* the suspension with 1.5 g/kg , twice a day for 31 days. The control group was administered the same volume of distilled water for 31 days. The daily dose of each drug was approximately 3 times the human equivalent dose, according to the dose recorded in the 2015 Chinese Pharmacopoeia. This dose was tested in the preliminary test.

The dried pulverized powders of RG, GS, GL and AG were separately sealed for preservation in brown bottles put in the fridge at a temperature of 4°C . Every single brown bottle contained two days' worth of doses and was suspended in water immediately before the drug administration each day.

2.5. Appearance change of the mice

The typical symptoms for the *fireness* of ginseng include increased eye gum, excitation, agitation, irritability, insomnia, dizziness in addition to other symptoms such as bleeding. Among them, eye gum is an obvious symptom. Thus, it was observed every day in the morning.

2.6. Measurement of the mice rectal temperature

Rectal temperature was measured with an electronic rectal thermometer at 1 h after the second drug administration of the day on the 10th, 15th, 20th, and 25th day of drug administration.

2.7. Measurement of mice saliva secretion

Differences in saliva secretion were also measured. Prior to the measurement, 5 mg of dry, aseptic, absorbent cotton ball was put into the 1.5 ml EP tube and then weighed accurately. On the 14th, 21th, and 28th day of drug administration, 1 h after the second drug administration of the day, the dry cotton ball in the EP tubes was put into the inner cheek of the mice mouth. Five minutes later, the wet cotton was taken out and weighed again. The weight of saliva secretion was calculated as: $(\text{mg}) = (\text{wet cotton weight} + \text{EP tube weight}) - (\text{dry cotton weight} + \text{EP tube weight})$.

2.8. Measurement of urine and manure of mice

One hour after the second administration of drugs in the day, each mouse was covered with a 500-ml beaker separately above a filter paper on the desk. Each mouse's urinary production was measured for 3 h, using the following calculation: urinary production (g) = weight of wet filter paper - weight of dry filter paper. Simultaneously, the manure of the mouse was collected and weighed.

2.9. Determination of swimming time of mice

On the 25th day of drug administration, a glass cylinder was filled with 20 cm of water in depth at a temperature maintained at $20 \pm 0.5^\circ\text{C}$. One hour after the last drug administration, one heavy lead which was approximately 10% of the body weight of the mouse was fastened on the tail of mouse. The mouse was then put into the glass cylinder for swimming. When the mouse's head submerged for 10 s in the water and could not rise to the water surface, the mouse was taken out of the water immediately. The time period was recorded as swimming time.

2.10. Determination of coagulation time

One hour after the last drug administration of the 30th day of drug administration, the mice were anesthetized with isoflurane. Then, blood was taken from the eye's orbital vein with a glass capillary of 1 mm inner diameter until the blood rose up the column to 8 cm in the glass capillary. Every 20 s, a section of the glass capillary tube was broken to check if there was the hemagglutination. The coagulation time was calculated from the time of blood collection into the glass capillary tube to the appearance of blood coagulation.

2.11. Determination of VO_2 , VCO_2 , H , RER and EE

The mice in the different groups were individually placed in the respiratory chambers of TSE phenoMaster/LabMaster (TSE phenoMaster/LabMaster, Germany) to monitor 24 h respiratory parameters after the drug or distilled water was administered on the

20th and 30th day at 8:00 am. Mice were continuously recorded for 24 h with the following measurements being taken every 30 min: water intake, food intake, and gas exchange (O_2 and CO_2) (using the TSE LabMaster system). VO_2 , VCO_2 , and energy expenditure were calculated according to the manufacturer's guidelines (PhenoMaster Software, TSE Systems). The respiratory exchange rate was estimated by calculating the ratio of VCO_2/VO_2 . Values were adjusted by body weight to the power of 0.75 ($kg^{-0.75}$) where mentioned. The room temperature was controlled at $22 \pm 1^\circ C$ in an isolated environment. The sampling flow rate was continuously controlled at 0.25 L/min, with the flow speed set at 0.35 L/min. This was continuously detected for 24 h to measure the VCO_2 , VO_2 , H, RER and EE of each group (Hofmann et al., 2007; Jing et al., 2013).

2.12. Effect on biochemical indexes of energy metabolism in serum and liver

2.12.1. Pyruvic acid in serum

Blood was collected from the eye orbital vein after isoflurane anesthesia, then cryogenically centrifuged at 3500 rpm/min for 10 min to isolate the serum. The pyruvic acid was then measured strictly according to the measurement kit.

2.12.2. SDH activity, LDH activity, Na^+K^+ -ATPase activity and glycogen in liver

The mice were sacrificed by the cervical dislocation method after anesthetizing with isoflurane and their livers were dissected rapidly. Approximately 0.2 g liver was homogenized with 9 times the amount of saline and then centrifuged at a speed of 2500 r/min for 10 min. The resulting supernatant was used for the measurement of SDH activity, LDH activity, and Na^+K^+ -ATPase activity according to correspondent measurement kits. About 75 mg of liver was weighed accurately for the determination of glycogen. All of the dissection parts of the liver were taken from the same position of the liver as much as possible, and it was refrigerated at $-80^\circ C$ before use (Wang and Zhao, 2007; Wang et al., 2013; Yin et al., 2012).

2.13. Statistical analysis

All data were expressed as the mean $\pm$ standard deviation with the use of SPSS19.0 software. The data were analysed using ANOVA with the one-way LSD test, unless otherwise specified, with Dunnett's T3 analysis. P-values of < 0.05 were considered to be statistically significant.

3. Results

3.1. Appearance change of mice

After 20 days of drug administration, the mice in the RG group exhibited a little excitation and their eye gum was obviously increased compared with the control and AG groups. The typical symptoms of mice on the 30th day of drug administration are shown in the Supplementary Material Fig. S-1.

3.2. The effects of rectal temperature in mice

The changes of mice rectal temperature for 25 days of drug administration showed (Fig. 1) that on the 10th day of drug administration, the rectal temperature of mice in the RG group was obviously increased compared with the control group. Additionally, the rectal temperature of mice in the GS group was markedly increased on the 15th day of drug administration

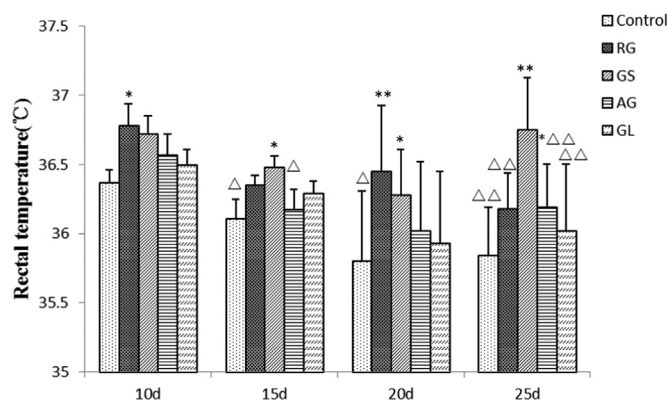


Fig. 1. The effects of RG, GS, AG and GL on rectal temperature after administrating for 10 days, 15 days, 20 days and 25 days. Cumulative values are reported as mean $\pm$ S. E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, ^ $P < 0.05$ and ^^ $P < 0.01$ compared to GS group, using ANOVA with the LSD analysis.

compared with the control group. On the 20th day of drug administration, the rectal temperature in the RG and GS groups were significantly higher than the control group ($P < 0.05$). On the 25th day of administration, the rectal temperature of mice in the GS group was remarkably increased compared with the control group, and it was also markedly higher than the AG and GL groups ($P < 0.01$).

3.3. The effects of saliva secretion in mice

The experiment results (Fig. 2) indicated that the saliva secretion of mice in the RG group, GS group and the GL group had no significant difference on the 21th day of drug administration compared with control group. On the 28th day of drug administration, the saliva secretion of mice in the RG group and GS group remarkably decreased compared with the control group ($P < 0.01$). Additionally, the saliva secretion in the GL group was also a bit lower than the control group ($P < 0.05$).

3.4. The effects on the excretion of urine and manure of mice

The excretion of urine and manure of the mice was measured on the 24th day of drug administration after the symptoms of *fireness* appeared in the RG and GS groups. As shown in Figs. 3 and 4, the results revealed that the amount of urine in the RG group,

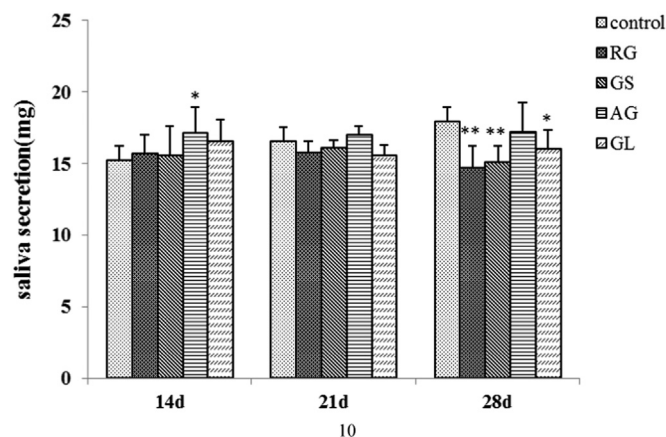


Fig. 2. The effects of GS, RG, AG and GL on saliva secretion after administrating for 14 days, 21 days and 28 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with the LSD analysis.

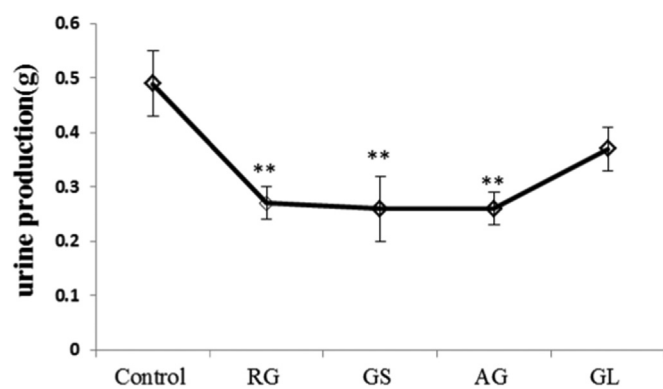


Fig. 3. The effects of GS, RG, AG and GL on urine production after administrating for 24 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with the LSD analysis.

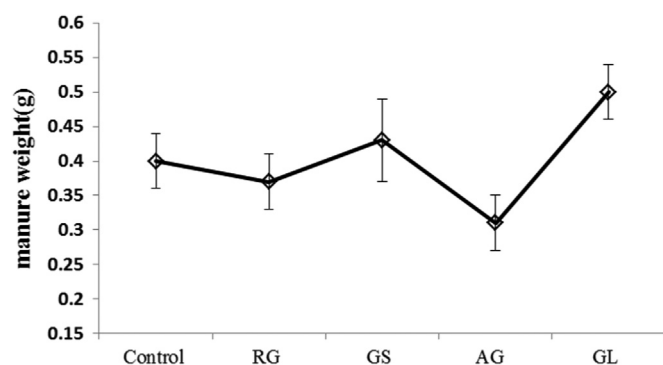


Fig. 4. The effects of GS, RG, AG and GL on manure production after administrated for 24 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with the LSD analysis.

GS group and AG group remarkably decreased compared with the control group. However, the amount of manure showed no significant difference among drug groups compared with the control group.

3.5. The effects on the swimming time in mice

As shown in Fig. 5, the results indicated that the swimming time of mice in GS group, GL group and AG group remarkably increased compared with the control group ($P < 0.01$) on the 25th

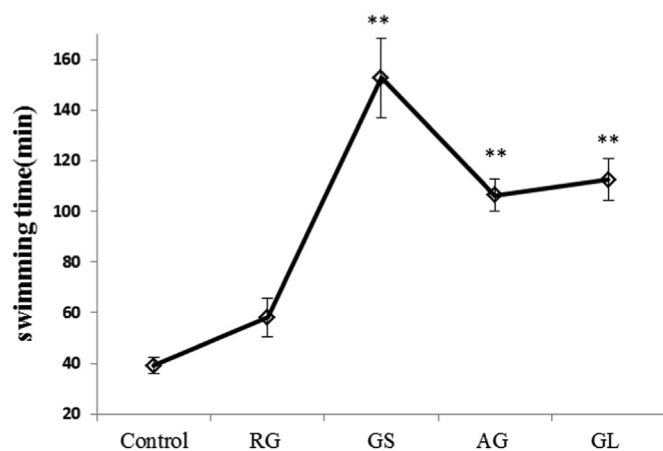


Fig. 5. The effects of GS, RG, AG and GL on swimming time after administrated for 25 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with Dunnett's T3 analysis.

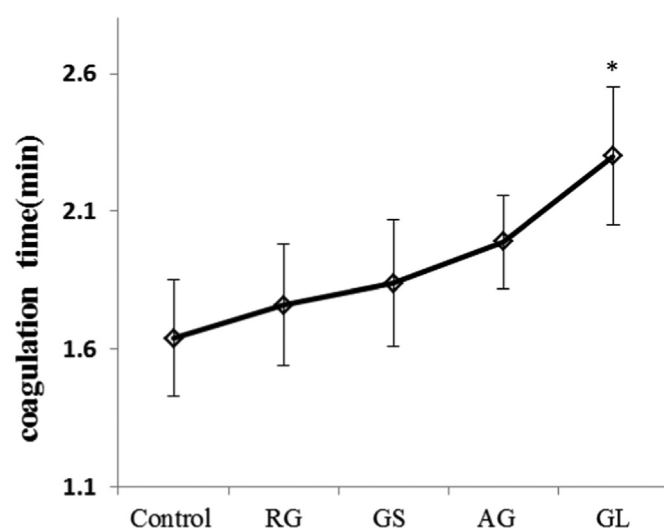


Fig. 6. The effects of RG, GS, AG and GL on coagulation time after administrating for 31 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with the LSD analysis.

day of drug administration.

3.6. The effects of coagulation time in mice

The results (Fig. 6) showed that the coagulation time in the GL group was evidently prolonged (compared with control group, $P < 0.05$). There were no obvious significant differences with other groups compared with control group on the 31st day of drug administration.

3.7. The effects of VO_2 , VCO_2 , H , RER and EE in mice

As shown in Figs. 7 and 8 (A, B, C, D), the data demonstrated that on the 20th day of administration, the VO_2 , VCO_2 , H and EE in every group showed no difference during the daytime. At night, the VCO_2 in the GS group was obviously decreased compared with the control group ($P < 0.05$). On the 30th day of administration, there were no significant changes in any group compared with the control group in the daytime. However, at night, the VO_2 , VCO_2 , and H in the ginseng group was obviously increased compared with the control group ($P < 0.05$). Additionally, EE in the GS group was significantly increased (compared with the control group, $P < 0.01$).

3.8. The effects on LDH activity in mice

Fig. 9 showed that LDH activity in each group had no obvious significant difference compared with control group, during 31 days of drug administration.

3.9. The effects of SDH activity in mice

As shown in Fig. 10, the results indicated that on the 21st day of administration, SDH activity in the GL group was distinctly decreased compared with the control group ($P < 0.05$). On the 31st day of administration, SDH activity in the AG group was obviously decreased ($P < 0.05$).

3.10. The effects of $Na^+-K^+-ATPase$ activity in mice

As shown in Fig. 11, the data demonstrated that on the 11th day of administration, $Na^+-K^+-ATPase$ activity in the GS group and the RG group distinctly increased compared with the control

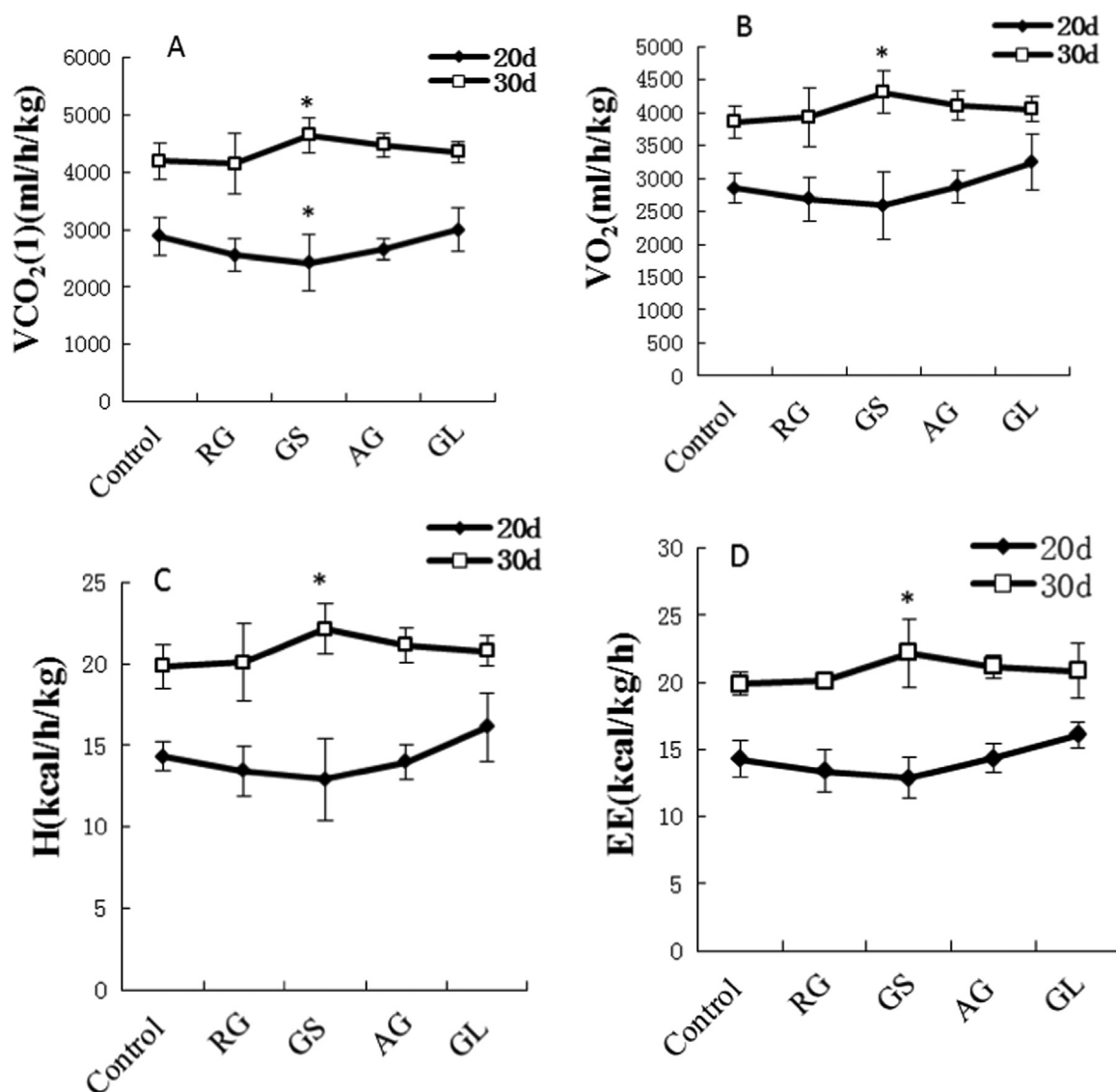


Fig. 7. (A, B, C, D) The effects of GS, RG, AG and GL on VCO₂ (ml/h/kg), VO₂ (ml/h/kg), H (kcal/h/kg), EE (kcal/kg/h) during the night after administrating for 20 days and 30 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with the LSD analysis.

group ($P < 0.05$). However, on the 21st day of administration, there was no obvious significant difference amongst the groups. On the 31st day of administration, Na⁺-K⁺-ATPase activity in the RG group was significantly decreased compared with the control group ($P < 0.01$).

3.11. The effects of pyruvic acid content in mice

The results (Fig. 12) showed that on the 11th day of administration, compared with the control group, the pyruvic acid content in the AG group and GL group was distinctly increased ($P < 0.05$). On the 21st day of administration, the pyruvic acid content in the RG group was obviously decreased ($P < 0.05$). On the 31st day of administration, the pyruvic acid content in the RG group was obviously increased (compared with the control group, $P < 0.01$).

3.12. The effects of liver glycogen content in mice

The results (Fig. 13) showed that liver glycogen content in the GS group and GL group was obviously increased compared with the control group ($P < 0.05$) on the 31st day of drug administration. The liver glycogen content in the AG group also showed an

increase, but there was no obvious significant difference compared with the control group.

4. Discussion

According to the theory of TCM, drugs with a warm nature could easily cause *dryness* (燥) and damage to *yin*, which leads to inner heat, i.e., *fire* (火), if used chronically at a high dosage. The first treatise on TCM summarized the characteristics of “dryness” as “excessive dryness that exhausts saliva and liquid secretions” (“燥胜则干” in Chinese). In Chinese folk lore, the experience with ginseng use is that chronic or high doses of ginseng can cause *fire*, which limits its widespread use. Clinically, the *fire* always exhibited a large amount of eye gum, excitation, dry and cracked lips, dry nose and throat, rough skin, dry hair and even bleeding. Thus, the rectal temperature, saliva secretion, coagulation time (Zhong et al., 2013), and excretion of urine and manure were always selected to indicate the dryness of drugs. In addition, the main efficacy of ginseng is its tonic action, so swimming time was selected to indicate its tonic effect. Up until now, there have been very few reports on the *fire* of ginseng. In TCM, the

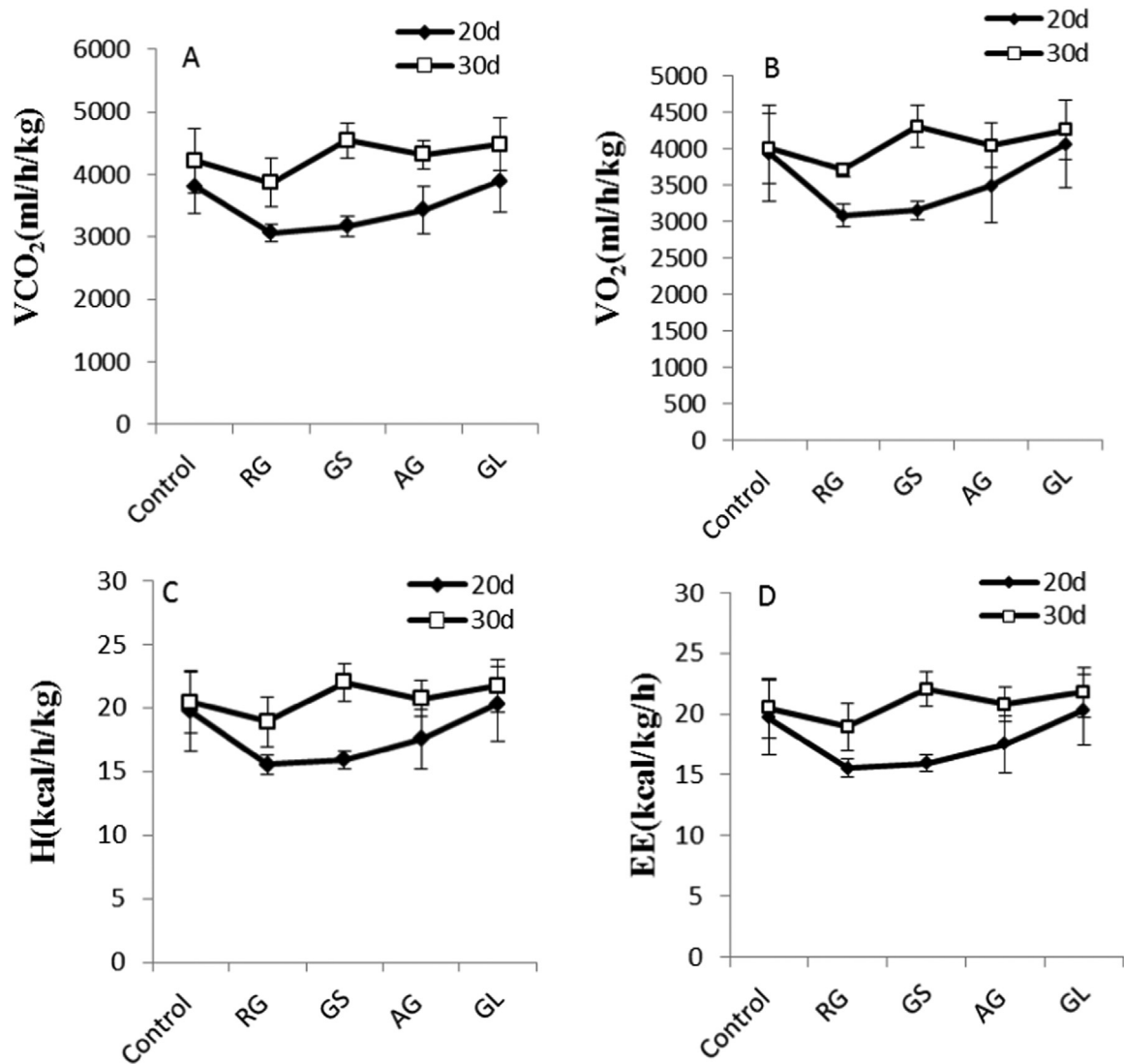


Fig. 8. (A, B, C, D) The effects of GS, RG, AG and GL on VCO₂ (ml/h/kg), VO₂ (ml/h/kg), H (kcal/h/kg), EE (kcal/kg/h) during the daytime after administering for 20 days and 30 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with the LSD analysis.

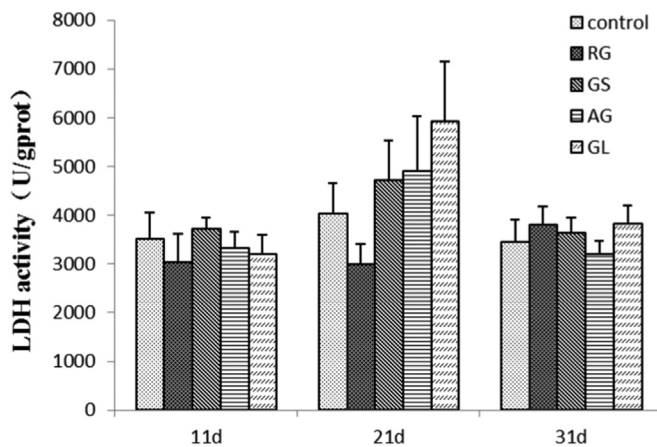


Fig. 9. The effects of RG, GS, AG and GL on LDH activity (U/gprot) in liver tissue after administering for 10 days, 21 days and 31 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with the LSD analysis.

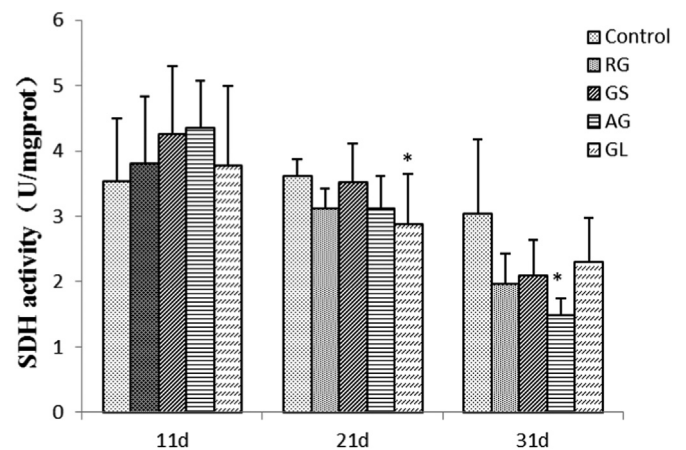


Fig. 10. The effects of RG, GS, AG and GL on SDH activity (U/mgprot) in liver tissue after administering for 11 days, 21 days and 31 days. Cumulative values are reported as mean $\pm$ S.E. for 10 rats in each group. * $P < 0.05$ and ** $P < 0.01$ compared to controls, using ANOVA with the LSD analysis.

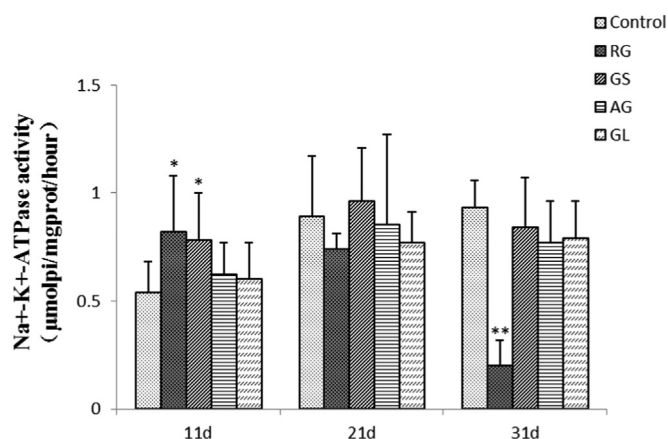


Fig. 11. The effects of RG, GS, AG and GL on Na⁺-K⁺-ATPase activity (μmolpi/mgprot/hour) in liver tissue after administrating for 11 days, 21 days and 31 days. Cumulative values are reported as mean ± S.E. for 10 rats in each group. *P < 0.05 and **P < 0.01 compared to controls, using ANOVA with the LSD analysis.

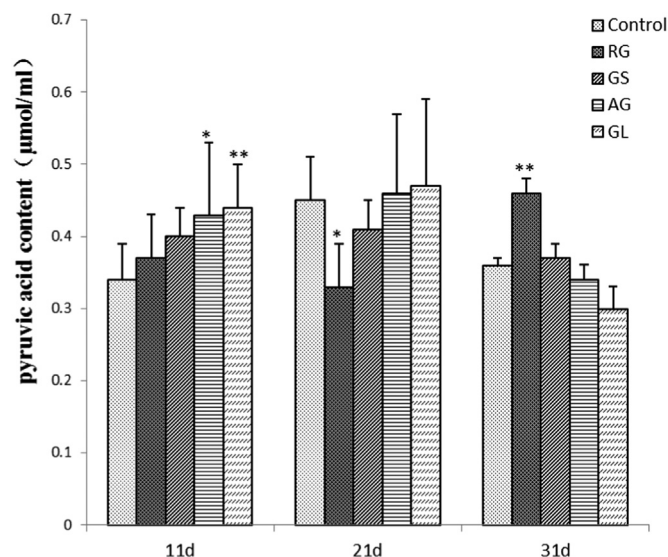


Fig. 12. The effects of RG, GS, AG and GL on pyruvic acid content (μmol/ml) in serum after administrating for 11 days, 21 days and 31 days. Cumulative values are reported as mean ± S.E. for 10 rats in each group. *P < 0.05 and **P < 0.01 compared to controls, using ANOVA with the LSD analysis.

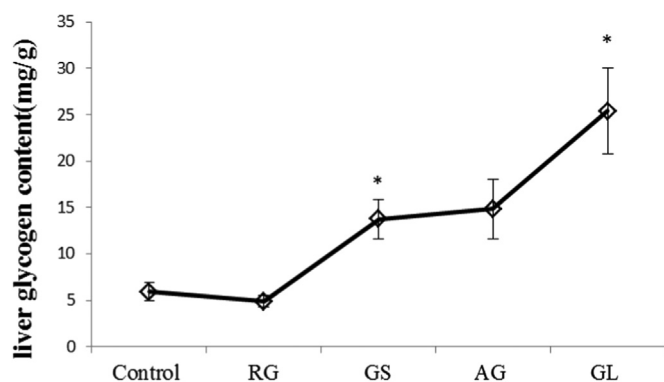


Fig. 13. The effects of GS, RG, AG and GL on liver glycogen content (mg/g) after administrating for 31 days. Cumulative values are reported as mean ± S.E. for 10 rats in each group. *P < 0.05 and **P < 0.01 compared to controls, using ANOVA with the Dunnett's T3 analysis.

natures of cold, cool, warm, and hot of drugs are mainly associated with energy metabolism (Wang et al., 2008). The current research indicates that warm drugs could enhance body energy metabolism, and cold drugs inhibit body energy metabolism (Huang et al., 2010a, 2010b, 2009). This shows that the drug effect on the body must be associated with energy conversion and transfer (Hu et al., 2008). For this reason, our experiment selected oxygen consumption (VO₂), carbon dioxide production (VCO₂), heat production (H) and energy expenditure (EE), measured by the TSE system, to measure the results. In addition, the pyruvic acid content in serum, Na⁺-K⁺-ATPase activity and the liver glycogen content were determined. According to China's Pharmacopoeia, RG, GS, AG and GL all possess different properties. The former two exhibit warm properties, with RG being warmer than GS. Although ginsenosides are their chief constituents, the proportions of ginsenosides in each are different. Therefore, the four drugs were chosen to study the dryness of ginseng according to each of their properties. Because the *fire*ness caused by ginseng requires chronic administration, the experiment was conducted over 31 days. To reflect the time dependence, the biochemical indices were measured constantly. The mice in the RG group exhibited excitation and showed an increasing amount of eye gum after 20 days of drug administration. Our experiments results indicated that warm-natured ginseng and red ginseng could distinctly increase the mouse rectal temperature, while reducing saliva secretion and urine volume at the same time, after 25 days drug administration. This indicates that the *fire*ness formed. The properties of AG and GL were cool and cold, and during the first 20 days of administration the rectal temperature had no change compared with the control group. In addition, the saliva secretion in the AG group increased significantly on the 14th day of administration, revealing that the *dryness* of AG and GL was not obvious.

The main pathways of energy metabolism are glycolysis and the citric acid cycle (TCA). Energy metabolism includes energy synthesis and decomposition. Succinate dehydrogenase (SDH), which is the only enzyme connected with intima, is an important metabolic enzyme in the tricarboxylic acid cycle and is mainly distributed in the inner mitochondrial membrane. Because the changes of SDH activity has been associated with the quantity and injury of mitochondria, it has often been regarded as a mitochondrial marker enzyme (Huang et al., 2005). The up-regulation of SDH activity marks the acceleration of TCA and the increase of ATP. SDH activity in the GL group decreased at the 21st day of administration, while SDH activity in the AG group reduced on the 31st day of administration. This indicates that AG and GL inhibited the activity of SDH by slowing down the TCA cycle to inhibit energy production. This fact corroborates with the reduction in the amount of urine and feces in AG group. It is also consistent with the literature, which reports that American ginseng total saponin had a strong sedatory function (Zhuang et al., 2003). These two groups had different inhibition times, which might be due to the cold property of GL and the cool property of AG. Therefore, the inhibition of energy metabolism in the GL group was stronger than AG group. Additionally, the timing of the appearance of the inhibitory effect of the GL group appeared earlier than for the AG group.

Lactate dehydrogenase (LDH) exists widely in the organization and its activity can usually reflect the level of anaerobic metabolism in skeletal muscle, heart, kidney and liver. Due to that, it is regarded as a marker for anaerobic metabolism enzymes (Liu, 2012). There were no differences among groups in our study with regards to LDH activity, indicating that there was no difference in the anaerobic metabolism.

ATPase exists in the cell or on the organelle membrane, playing an important role in the aspects of substance transport, information transfer and energy conversion. The Na⁺-K⁺-ATP enzyme is

an important enzyme that balances the ion concentration of the two sides of the membrane through the consumption of ATP. As the enzyme activity increases, the energy consumption and the heat production increases (Sudar et al., 2008). Compared with the control group, Na^+/K^+ -ATPase activity was significantly increased when RG and GS were administered for 11 days. Na^+/K^+ -ATPase activity of the RG group decreased but did not show a significant difference after 21 days of administration. On the 31st day of administration, Na^+/K^+ -ATPase activity significantly decreased in the RG group, which is consistent with the warm property of GS and RG. So, in the early days of drug administration, they could enhance the Na^+/K^+ -ATPase activity, and accelerate the release of ATP. However, along with increasing days of exposure, the body produced large amounts of energy to form the *fireness*. To reduce the excessive energy consumption and balance the body's energy metabolism, the energy metabolism was inhibited. This result was similar to the study that showed the activity of liver Na^+/K^+ -ATPase decreased significantly in the rat model of wet and hot symptoms (Cheng et al., 2007).

Pyruvate is the final product of glycolysis, which occupies a pivotal role in glycolysis and the TCA. Glycolysis is a common process in the aerobic respiration and anaerobic respiration pathway. In aerobic conditions, pyruvic acid could generate acetylCoA through oxidative decarboxylation, which then enters the TCA, where it is eventually completely oxidized into CO_2 , H_2O and large amounts of ATP. Under anaerobic conditions, hydrogenation reduces the lactic acid, and the increasing amount of pyruvate content could accelerate the glucose metabolism and the enhancement of energy metabolism of body. The pyruvate content of mice in the RG group increased a little compared with the control group when it was administered for 11 days. On the 21st day of administration, it was clearly reduced, but then significantly increased on the 31st day of administration. It could be concluded that RG could increase the rate of glycolysis, increasing the pyruvate content, in the early days. Then, pyruvate was oxidized to acetylCoA and participated in the TCA, which decreased the pyruvic acid content significantly. After that, pyruvate content increased again, and it could be speculated by the dynamic process of glucose metabolism that the red ginseng may accelerate energy supply through the decomposition of sugar. The GS group showed the same process as the red ginseng, but there was no significant differences. This may be because RG was far more warm-natured than ginseng, so its increase in energy metabolism was stronger than ginseng. On the 11th day of administration, pyruvate content in the AG and GL groups increased obviously, but then later gradually decreased. This may be because of the rapid speed of glycolysis of these two groups in the early days, while later, their cold properties showed the inhibitory effect on energy metabolism.

Glycogen is a polysaccharide polymerized by large number of glucose molecules and is stored in the liver in the form of glycogen. It could be broken down into glucose when required, simultaneously producing energy. GS, GL and AG groups all exhibited anti-fatigue action as demonstrated by swimming times. On the 25th day of drug administration, the swimming times of the GS group, AG group and the GL group were significantly longer compared with the control group. Simultaneously, the liver glycogen content of the three groups increased significantly, revealing that the anti-fatigue action of GS, AG and GL was likely to play a role by increasing the preservation of liver glycogen. Our results were consistent with the report that GS's anti-fatigue action is stronger than RG's in the literature (Gao et al., 2011). The glucose can metabolize pyruvic acid in the aerobic respiration and anaerobic respiration pathway. Glycogen is a type of polysaccharide which is stored in the liver as a form of energy. Liver glycogen can decompose to glucose molecules, which then generates pyruvate when the body requires energy. Hence, liver glycogen decreases

along with the increasing pyruvate. On the 31st day of RG administration, the liver glycogen decreased along with the increasing pyruvate. This indicates that RG could accelerate the rate of glycolysis, which decreased liver glycogen. Pyruvate is the final product of glycolysis, so the pyruvate content in the serum increased. Thus, the liver glycogen decreasing along with the increasing pyruvate is mainly associated with the process of glycogenolysis.

Compared with the control group, VCO_2 , H and EE in the GS group obviously increased when administered for 31 days. It could be speculated that GS could accelerate the generation, storage and use of the energy. However, the RG group showed no statistical significance when compared with the control group, which may be because red ginseng is far more warm-natured than ginseng. On the 31st day of administration, the body showed some dryness. To reduce the excessive energy consumption and maintain the normal metabolism of the human body, red ginseng showed a certain role in the inhibition of energy metabolism to normalize the body's metabolism.

Our results indicate that the rectal temperature significantly increased, while saliva secretion and urine volume decreased significantly in the RG and GS groups. The H also significantly increased in GS group, indicating that the *fireness* has been formed in RG group. The *fireness* in cold and cool-natured AG and GL was not obvious, so AG and GL could obviously inhibit the body energy consumption. Moreover, it is consistent with the TCM opinion that the basal metabolic rate of cold syndrome patients is lower, while the basal metabolic rate of heat syndrome patients is higher. Our results indicated that *fireness* of warm-natured RG and GS may be closely related to the increasing of energy metabolism. Both of them could be enhancing energy metabolism to accelerate the utilization of energy to produce the heat. Then, after a prolonged phase of energy consumption, *fireness* is produced. This is the first time that the *fireness* of ginseng has been closely associated with energy metabolism. It is also the first time that it has been shown that Na^+/K^+ -ATPase activity is largely inhibited and pyruvic acid obviously increased in the state of *fireness*. Because pyruvic acid is only one type of intermediate metabolite of energy metabolism, Na^+/K^+ -ATPase activity was regarded as the main index to reflect the *fireness* of ginseng.

5. Conclusion

In summary, we have demonstrated that the *fireness* of ginseng is closely associated with energy metabolism. The warm medicines RG and GS possibly enhance the body's energy metabolism, therefore increasing heat production and leading to the dryness. GL and AG, with cold properties, could inhibit the body's energy metabolism. Therefore, their dryness is not obvious.

Author contributions statement

Deqiang Dou designed and guided the experiments, Xu Xu and Deqiang Dou conducted the experiments and analysed the measured data, Deqiang Dou made the final conclusion. Both authors reviewed the manuscript.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.jep.2016.06.024>.

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